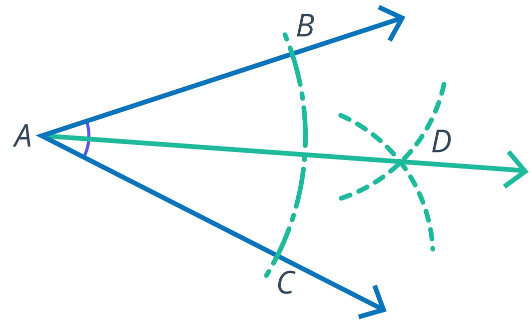
Justifying constructions using congru[redacted]riangles

1 Consider the construction that has been created so that the arcs with centers at B and C are congruent.

Describe $\overrightarrow{AD}$.

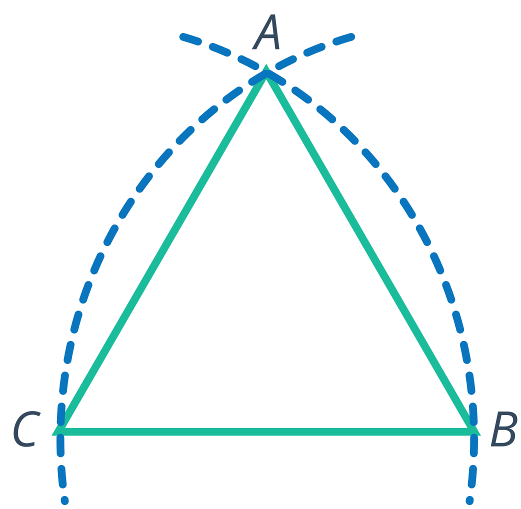


2 Consider the arc of the circle centred at C that passes through B and the arc of the circle centred at B passes through C. Both arcs intercept at a point A, and the triangle ABC is constructed as shown:

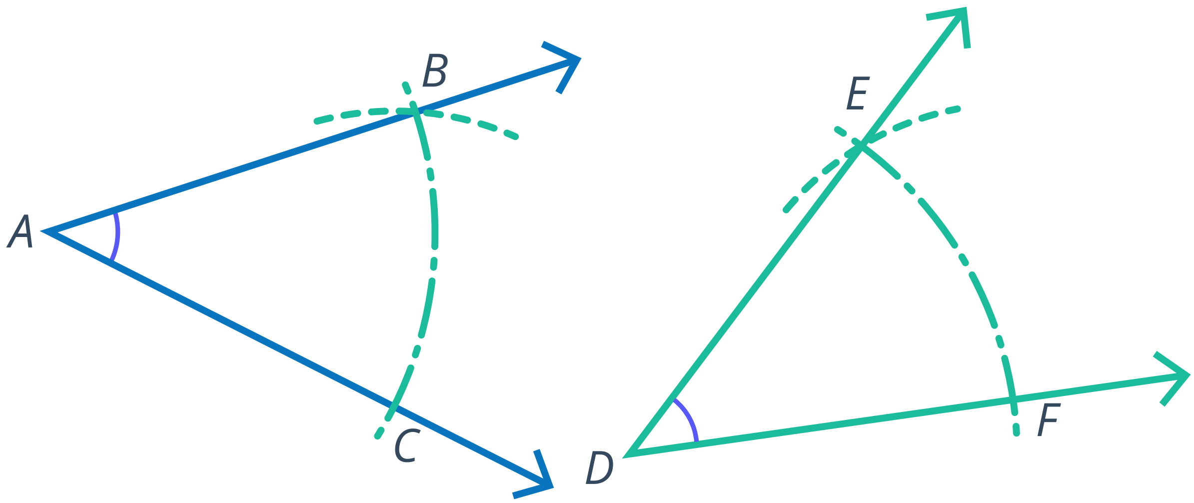
a Write line segments that are congruent radii of the circle centred at C.

b Write the line segments that are congruent radii of the circle centred at B.

c Classify the triangle ABC as scalene or equilateral.



3 Consider the angle $\angle EDF$, constructed fro[redacted] given angle $\angle BAC$ by drawing a set of dotted arcs.



It can be determined from the construction that:

- $\overline{AB} \cong \overline{DE}$
- $\overline{AC} \cong \overline{DF}$
- $\overline{BC} \cong \overline{EF}$

a State the reason that the line segments above are congruent.

b Are the triangles $\triangle ABC$ and $\triangle DEF$ congruent? If yes, state the congruence theorem that justifies the congruence.

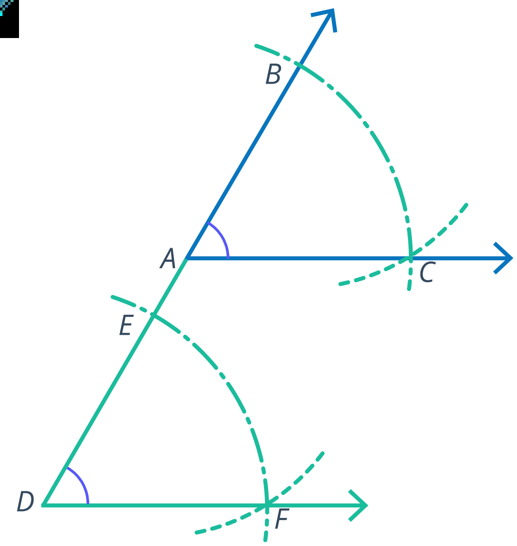
c Determine whether the following reasons prove that $\angle BAC$ and $\angle EDF$ are congruent.

i It can be shown that the angles are corresponding parts of congruent triangles.

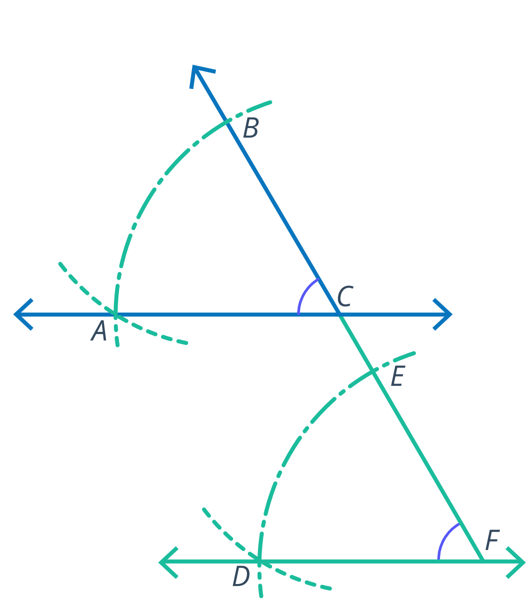
ii The angles were measured with a protractor and shown to have approximately the same measure.

iii The angles look like they are the same size.

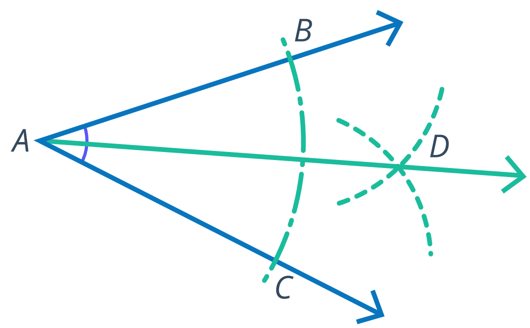
4 Consider the given diagram, where $\angle EDF$ is constructed congruent to $\angle BAC$. Justify the steps of construction in a two-column proof.



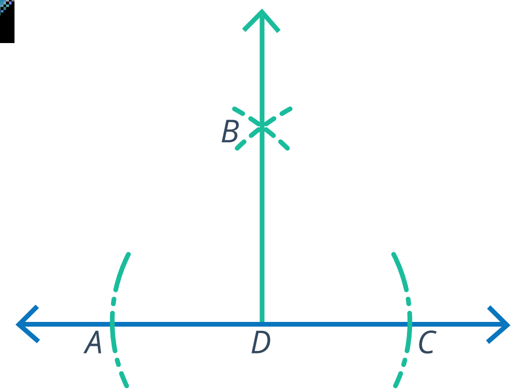
5 Consider the given diagram, where $\overleftrightarrow{DF}$ is constructed parallel to $\overleftrightarrow{AC}$. Justify the steps of construction in a two-column proof.



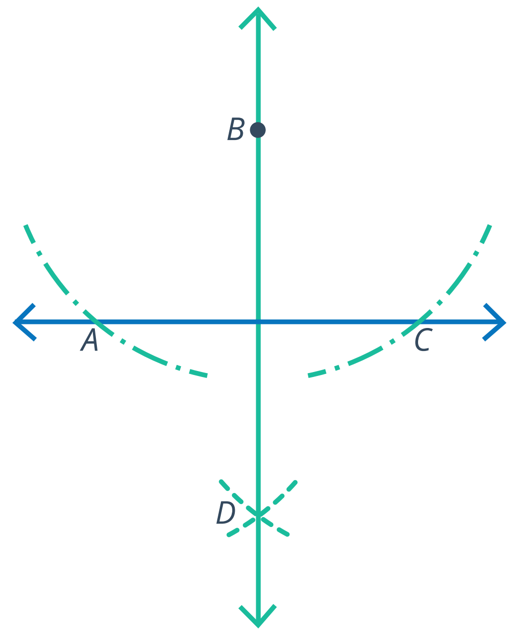
6 Consider the given diagram:
Prove that the construction of $\overrightarrow{AD}$ is an angle bisector of the given angle $\angle BAC$.



7 Consider the given diagram, where $\overrightarrow{DB}$ is constructed perpendicular to $\overleftrightarrow{AC}$. Justify the steps of construction in a two-column proof.



8 Consider the given diagram, where $\overleftrightarrow{BD}$ is constructed perpendicular to $\overleftrightarrow{AC}$. Justify the steps of construction in a two-column proof.



9 Consider the given diagram.

a $\overleftrightarrow{CD}$ is constructed from the given line $\overleftrightarrow{AB}$ by drawing a set of dotted arcs. Prove that the point E is the midpoint of $\overline{AB}$.

b $\overleftrightarrow{CD}$ is constructed so that it is a perpendicular bisector of $\overline{AB}$ and $\overleftrightarrow{AB}$ is a perpendicular bisector of $\overline{CD}$. Justify the steps of construction in a two-column proof.